

# Bisection Method

July 2026

## 1 Introduction

The Bisection Method is a numerical root-finding algorithm used to find a solution of the nonlinear equation

$$f(x) = 0.$$

It is one of the simplest and most reliable root-finding methods because it only requires the function to be continuous on an interval where the function values at the endpoints have opposite signs.

## 2 Mathematical Foundation

### 2.1 Intermediate Value Theorem

If  $f$  is continuous on a closed interval  $[a, b]$ , and  $c$  is any number between  $f(a)$  and  $f(b)$  inclusive, then there is at least one number  $x$  in the closed interval such that  $f(x) = c$ .

### 2.2 Bisection Method

The Bisection Method is based on the Intermediate Value Theorem. Let  $f$  be continuous on the interval  $[a, b]$ . If  $f(a)f(b) < 0$ , then there exists at least one point  $c \in [a, b]$  such that  $f(c) = 0$ .

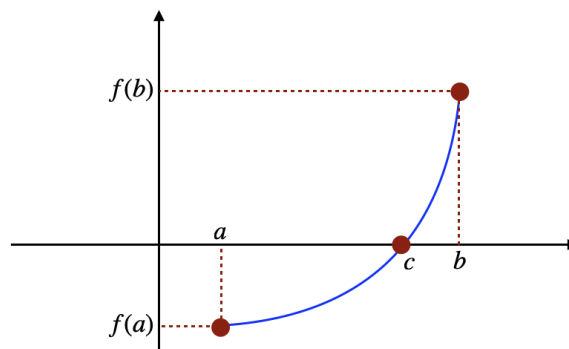


Figure 1: Bisection Method

## 3 Algorithm Flow

At every iteration, the current interval is divided into two equal sub-intervals. The sub-interval that still contains a sign change is guaranteed to contain at least one root. Therefore, the interval containing the root becomes smaller after every iteration.

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**Algorithm 1** Bisection Method

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**Input:** Continuous function  $f(x)$ , interval  $[a, b]$

**Output:** Approximate root  $m$

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if  $f(a)f(b) \geq 0$  then
    return failure
end if
while not converged do
     $m = a + (b - a)/2$ 
    if  $f(a)f(m) < 0$  then
         $b = m$ 
    else
         $a = m$ 
    end if
end while
return  $m$ 
```

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## 4 Convergence Analysis

### 4.1 Interval Reduction

At each iteration, the Bisection Method divides the current interval into two equal halves. Therefore, the interval length is reduced by a factor of two after every iteration.

Let the initial interval length be

$$L_0 = b - a.$$

Since the interval is halved at every iteration,

$$L_1 = \frac{L_0}{2},$$

$$L_2 = \frac{L_0}{2^2},$$

$$\vdots$$

Hence,

$$L_n = \frac{L_0}{2^n} = \frac{b - a}{2^n}.$$

### 4.2 Required Iterations

Suppose the desired accuracy is  $\varepsilon$ . Since the interval length after  $n$  iterations is

$$L_n = \frac{b - a}{2^n},$$

the stopping criterion requires

$$\frac{b - a}{2^n} \leq \varepsilon.$$

Multiplying both sides by  $2^n$  gives

$$b - a \leq \varepsilon 2^n.$$

Dividing both sides by  $\varepsilon$ ,

$$\frac{b-a}{\varepsilon} \leq 2^n.$$

Taking the base-2 logarithm of both sides,

$$\log_2 \left( \frac{b-a}{\varepsilon} \right) \leq n.$$

Therefore, the minimum number of iterations required is

$$n \geq \log_2 \left( \frac{b-a}{\varepsilon} \right)$$

This result shows that the number of iterations grows logarithmically with respect to the desired accuracy. Consequently, obtaining higher accuracy requires only a moderate increase in the number of iterations.

### 4.3 Error Bound

Let  $x^*$  denote the exact root and let  $m_n$  be the midpoint after the  $n$ th iteration.

Since the exact root always remains inside the current interval, the maximum error is bounded by half of the interval length:

$$|x^* - m_n| \leq \frac{L_n}{2}.$$

Substituting the interval length

$$L_n = \frac{b-a}{2^n},$$

gives

$$|x^* - m_n| \leq \frac{b-a}{2^{n+1}}.$$

Therefore, the approximation error decreases exponentially as the number of iterations increases.

### 4.4 Convergence Rate

The Bisection Method has **linear convergence**. The error is reduced by approximately one-half at each iteration. However, it is guaranteed to converge provided that

- the function is continuous on  $[a, b]$ , and
- $f(a)f(b) < 0$ .

For this reason, the Bisection Method is regarded as one of the most robust root-finding algorithms.

## 5 Summary

The convergence properties of the Bisection Method can be summarized as follows:

- Interval length after  $n$  iterations:

$$L_n = \frac{b-a}{2^n}$$

- Required iterations:

$$n \geq \log_2 \left( \frac{b-a}{\varepsilon} \right)$$

- Error bound:

$$|x^* - m_n| \leq \frac{b-a}{2^{n+1}}$$

- Convergence order:

Linear Convergence

## 6 Numerical Example

To illustrate the Bisection Method, consider the nonlinear equation

$$f(x) = x^2 - 2.$$

The exact root is

$$x = \sqrt{2} \approx 1.41421356.$$

Choose the initial interval

$$[a, b] = [1, 2],$$

Since  $f(1)f(2) < 0$ , the Intermediate Value Theorem guarantees that at least one root exists in the interval  $[1, 2]$ .

| Table 1: $f(x) = x^2 - 2$ |          |                           |
|---------------------------|----------|---------------------------|
| Iteration                 | $m$      | $f(m)$                    |
| 1                         | 1.5000   | 0.2500                    |
| 2                         | 1.2500   | -0.4375                   |
| 3                         | 1.3750   | -0.1094                   |
| 4                         | 1.4375   | 0.0664                    |
| $\vdots$                  | $\vdots$ | $\vdots$                  |
| 27                        | 1.41421  | $-5.23681 \times 10^{-9}$ |

After 27 iterations, the approximate root is 1.41421.